Surjya Ghosh

D - 242, Dept. of Computer Science and Information Systems
BITS Pilani K K Birla Goa Campus, NH 17B, Zuarinagar, Goa - 403726
https://surjya-ghosh.github.io/ surjyag@goa.bits-pilani.ac.in

RESEARCH INTERESTS Affective Computing, Applied AI, Human Computer Interaction.

WORK EXPERIENCE

Associate Professor Jul 2026 - to date.
- *BITS Pilani K K Birla Goa Campus*, Goa, India
- Dept: Computer Science & Information Systems

Assistant Professor Aug 2021 - Jun 2026
- *BITS Pilani K K Birla Goa Campus*, Goa, India
- Dept: Computer Science & Information Systems

Postdoctoral Researcher Aug 2019 - Jul 2021
- *Centrum Wiskunde & Informatica*, Amsterdam, The Netherlands
- Group: Distributed & Interactive Systems

SAP CRM Manager Jul 2014 - Jan 2015
- *Capgemini India*, Kolkata, India
- Led the SAP CRM Implementation of SAB Miller Account.

Senior Advisory Consultant Mar 2013 - Mar 2014
- *IBM India*, Kolkata, India
- Worked as a SAP CRM Functional Consultant for Welch Allyn project.

Assistant Consultant Oct 2003 - Mar 2013
- *TATA Consultancy Services Ltd.*, India
- Worked as a SAP CRM Functional Consultant for AGL Energy in Australia.
- Worked as a Business Analyst for CitiGroup, Eli Lilly and Company in USA.

EDUCATION

Doctor of Philosophy (Ph.D.) Jan 2015 - Jul 2019
- Indian Institute of Technology Kharagpur, WB, India.
- Computer Science & Engineering
- Advisors: Dr. Bivas Mitra & Prof. Niloy Ganguly
- Thesis: Developing Smartphone Keyboard Interaction-based Emotion Detection System

Master of Technology (M.Tech.) 2012 - 2014
- Indian Institute of Technology Kharagpur, WB, India.
- Information and Communication Technology
- CGPA: 9.53/10
- Thesis: Dynamic Community Detection In Evolutionary Network.
- Ranked 1st in the department.

Bachelor of Technology (B.Tech.) 1999 - 2003
- Haldia Institute of Technology, Vidyasagar University, WB, India.
- Computer Science & Engineering
- Total Marks: 87.5%
- Thesis: Workflow Implementation in Library Management System.
- Ranked 2nd (silver medal) in the university.

Higher Secondary Examination (10 + 2) 1999
- West Bengal Council of Higher Secondary Education, India
- Total Marks: 88.9%

Secondary Examination (10) 1997
- West Bengal Board of Secondary Education, India
- Total Marks: 87.25%

Key Achievements

- Received Alexander von Humboldt Foundation Fellowship for CONNECT Follow-up program (2025).
- Received Unrestricted Research Gift 30K USD (for research on Indian Sign Language Recognition) from Google (2025, 2024).
- Best Runner up WIP paper at IEEE PerCom 2025.
- Selected as young researcher for 8th Heidelberg Laureate Forum (HLF) 2021.
- Best poster award (3rd Place) at COMSNETS 2019.
- Best paper award at IEEE ISCC Workshops - ICTS4eHealth 2018.
- Ranked 1st in M.Tech.(Information & Communication Technology), IIT Kharagpur, India.
- Received University Silver Medal for 2nd rank in B.Tech.(Computer Science & Engineering), Vidyasagar University, India.

Publications

DBLP author profile: https://dblp.org/pers/g/Ghosh:Surjya.html
Google Scholar profile: https://scholar.google.com/citations?user=jwQqy80AAAAJ
Total number of citations (as of 03rd Nov, 2026): 715, h-index: 13.

Articles in Books

[1] **Surjya Ghosh**, Johanna Lochner, Bivas Mitra, and Pradipta De, "Your Smartphone Knows You Better Than You May Think: Emotional Assessment 'On the Go' via TapSense", Edited book volume Quantifying Quality of Life: Incorporating Daily Life into Medicine, Katarzyna Wac and Sharon Wulfovich (eds.), Springer, 2022, 209–267. DOI:10.1007/978-3-030-94212-0_10.

Articles in Peer-reviewed Journals

[2] Panyu Zhang, Minseo Park, Tomiris Ismatzoda, Azizbek Mustafakulov, Uzair Ahmed, Otabek Najimov, Jumabek Alikhanov, **Surjya Ghosh**, and Uichin Lee, "CrossShift: Quantifying Interpersonal Differences in Mobile Sensing for Mental Health", Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies **(ACM IMWUT/UbiComp)**, Vol 10(3), 2026. DOI:10.1145/3831954. **(Impact Factor: 6.2)**

[3] Kshitish Ghate, Hrithik Nambiar, Alfiya M. Shaikh, Swarnali Banik, **Surjya Ghosh**, Niloy Ganguly, and Snehanshu Saha, "SLAB: A Self-supervised Label Generation Framework to Reduce Annotation Overhead", IEEE Transactions on Affective Computing, Vol 17(3), 2026, pp. 3389 - 3404. DOI:10.1109/TAFFC.2026.3685365. **(Impact Factor: 11.3)**

[4] Anuja Pinge, Vinaya Gad, Dheryta Jaisighani, **Surjya Ghosh**, and Sougata Sen, "Detection and monitoring of stress using wearables: a systematic review", Frontiers in Computer Science **(Mobile and Ubiquitous Computing)**, Vol 6, 2024. DOI:10.3389/fcomp.2024.1478851. **(Impact Factor: 3.4)**

[5] Swarnali Banik, Sougata Sen, Snehanshu Saha, and **Surjya Ghosh**, "Towards Reducing Continuous Emotion Annotation Effort During Video Consumption: A Physiological Response Profiling Approach", Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies **(ACM IMWUT/UbiComp)**, Vol 8(3), 2024. DOI:10.1145/3678569. **(Impact Factor: 6.2)**

[6] M. Prajwal, Ayush Raj, Sougata Sen, Snehanshu Saha, and **Surjya Ghosh**, "Towards Efficient Emotion Self-report Collection Using Human-AI Collaboration: A Case Study on Smartphone Keyboard Interaction", Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies **(ACM IMWUT/UbiComp)**, Vol 7(2), 2023. DOI:10.1145/3596269. **(Impact Factor: 6.2)**

[7] Shruti Rao, **Surjya Ghosh**, Gerard Pons Rodriguez, Thomas Röggla, Pablo Cesar, and Abdallah El Ali, "From Video to Hybrid Simulator: Exploring Affective Responses toward Non-Verbal Pedestrian Crossing Actions using Camera and Physiological Sensors", International Journal of Human-Computer Interaction, 2023. DOI:10.1080/10447318.2023.2224955. **(Impact Factor: 6.1)**

[8] **Surjya Ghosh**, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Designing An Experience Sampling Method for Smartphone based Emotion Detection", IEEE Transactions on Affective Computing, Vol 12(4), 2021, pp. 913 - 927. DOI:10.1109/TAFFC.2019.2905561. **(Impact Factor: 11.3)**

[9] **Surjya Ghosh**, Kaustubh Hiware, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Emotion Detection from Touch Interactions during Text Entry on Smartphones", International Journal of Human-Computer Studies, Elsevier, Vol 130, 2019, pp 47 - 57. DOI:10.1016/j.ijhcs.2019.04.005. **(Impact Factor: 6.0)**

Articles in Peer-reviewed Conferences

[10] Sanskriti Uma, **Surjya Ghosh**, and Dio Dzaky Achmad Mustaqim, "Smarter Together: Enhancing Human-AI Collaborative Grading With Teacher-Cognition Multi-Agent LLM Framework", The Annual ACM Conference on Intelligent User Interfaces (**ACM IUI 2026**), Paphos, Cyprus. DOI:10.1145/3742413.3789130.

[11] **Surjya Ghosh**, and Natalie Packham, "Towards Understanding Financial Decision-Making in Equity Markets Among German and Indian Participants: Survey-based Findings", Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (**ACM CHI EA 2026**), Barcelona, Spain. DOI:10.1145/3772363.379897.

[12] Swarnali Banik, Sougata Sen, Snehanshu Saha, and **Surjya Ghosh**, "Influence of Demographics and Personality Traits on Physiological Responses to Improve Continuous Emotion Annotation in Video Applications", Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (**ACM CHI EA 2025**), Yokohama, Japan. DOI:10.1145/3706599.3720095.

[13] Anuja Achyut Pinge, Amey Damle, Rishav Mukherji, Bhargav Nagaraj, **Surjya Ghosh** and Sougata Sen, "WhoAmIOn: A Technique To Determine Whether All Devices Are Being Used By The Same Person", 21st IEEE International Conference on Ubiquitous Intelligence and Computing (**IEEE UIC 2024**), Denarau Island, Fiji. DOI:10.1109/SWC62898.2024.00155.

[14] Swarnali Banik, Sougata Sen, Snehanshu Saha and **Surjya Ghosh**, "Improving Continuous Emotion Annotation in Video Platforms via Physiological Response Profiling", 12th International Conference on Affective Computing & Intelligent Interaction (**ACII 2024**), Glasgow, UK. DOI:10.1109/ACII63134.2024.00047.

[15] Alfiya M. Shaikh, Hrithik Nambiar, Kshitish Ghate, Swarnali Banik, Sougata Sen, **Surjya Ghosh**, Vaskar Raychoudhury, Niloy Ganguly, and Snehanshu Saha, "Self-SLAM: A Self-Supervised Learning Based Annotation Method to Reduce Labeling Overhead", European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML PKDD 2024**), Vilnius, Lithuania. DOI:10.1007/978-3-031-70378-2_8.

[16] **Surjya Ghosh**, Salma Mandi, Sougata Sen, Bivas Mitra, and Pradipta De, "Towards Estimating Missing Emotion Self-reports Leveraging User Similarity: A Multi-task Learning Approach", 2024 CHI Conference on Human Factors in Computing Systems (**ACM CHI 2024**), Hawai'i, USA. DOI:10.1145/3613904.3642833.

[17] Salma Mandi, **Surjya Ghosh**, Pradipta De, and Bivas Mitra, "SELFI: Evaluation of Techniques to Reduce Self-report Fatigue by Using Facial Expression of Emotion", 19th International Conference of Technical Committee 13 (Human- Computer Interaction) of IFIP (**INTERACT 2023**), York, UK. DOI:10.1007/978-3-031-42280-5_39.

[18] Satchit Hari, Ajay N, Sayan Sarcar, Sougata Sen, and **Surjya Ghosh**, "AffectPro: Towards Constructing Affective Profile Combining Smartphone Typing Interaction and Emotion Self-reporting Pattern", 24th ACM International Conference on Multimodal Interaction (**ACM ICMI 2022**), Bangalore, India. DOI:10.1145/3536221.3556603.

[19] **Surjya Ghosh**, Bivas Mitra, and Pradipta De, "ALOE: Active Learning based Opportunistic Experience Sampling for Smartphone Keyboard driven Emotion Self-report Collection", 10th International Conference on Affective Computing & Intelligent Interaction (**ACII 2022**), Nara, Japan. DOI:10.1109/ACII55700.2022.9953819.

[20] Shruti Rao, **Surjya Ghosh**, Gerard Pons Rodriguez, Thomas Röggla, Abdallah El Ali, and Pablo Cesar, "Investigating Affective Responses toward In-Video Pedestrian Crossing Actions using Camera and Physiological Sensors", 14th International ACM Conference on Automotive User Interfaces and Interactive Vehicular Applications (**ACM AutomotiveUI 2022**), Seoul, South Korea. DOI:10.1145/3543174.3546842.

[21] **Surjya Ghosh**, Gerard Pons, Shruti Rao, Abdallah El Ali, and Pablo Cesar, "Exploring Emotion Responses toward Pedestrian Crossing Actions for Designing In-vehicle Empathic Interfaces", CHI Conference on Human Factors in Computing Systems Extended Abstracts (**ACM CHI EA 2022**), New Orleans, USA. DOI:10.1145/3491101.3519764.

[22] Salma Mandi, **Surjya Ghosh**, Pradipta De, and Bivas Mitra, "Emotion Detection from Smartphone Keyboard Interactions: Role of Temporal vs Spectral Features", The 37th ACM/SIGAPP Symposium On Applied Computing (**ACM SAC 2022**), Brno, Czech Republic. DOI:10.1145/3477314.3507159.

[23] Tejal Karnavat, Jaskaran Bhatia, **Surjya Ghosh**, and Sougata Sen, "Exploring the challenges of using food journaling apps: A case-study with young adults", 18th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (**MobiQuitous 2021**), Beppu, Japan. pp 57 - 83. DOI:10.1007/978-3-030-94822-1_4.

[24] **Surjya Ghosh**, and Tanaya Guha, "Towards Autism Screening through Emotion-guided Eye Gaze Response", 43rd International Conference on Engineering in Medicine and Biology (**IEEE EMBC 2021**), Virtual. pp 820 - 823. DOI:10.1109/EMBC46164.2021.9630888.

[25] **Surjya Ghosh**, Salma Mandi, Bivas Mitra, and Pradipta De, "Exploring Smartphone Keyboard Interactions for Experience Sampling Method driven Probe Generation", 26th International Conference on Intelligent User Interfaces (**ACM IUI 2021**), Texas, USA. pp. 144-149. DOI:10.1145/3397481.3450669.

[26] **Surjya Ghosh**, Bivas Mitra, and Pradipta De, "Towards Improving Emotion Self-report Collection using Self-reflection", Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (**ACM CHI EA 2020**), Honolulu, USA. pp. 1 - 8. DOI:10.1145/3334480.3383019.

[27] Tong Xue, **Surjya Ghosh**, Gangyi Ding, Abdallah El Ali, and Pablo Cesar, "Designing Real-time, Continuous Emotion Annotation Techniques for 360° VR Videos", Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (**ACM CHI EA 2020**), Honolulu, USA. pp. 1 - 9. DOI:10.1145/3334480.3382895.

[28] Soumyajit Chatterjee, Adrija Bhowmik, Arun Singh, **Surjya Ghosh**, Bivas Mitra, and Sandip Chakraborty, "Detecting Mobility Context over Smartphones using Typing

and Smartphone Engagement Patterns", 18th IEEE International Conference on Pervasive Computing & Communications (**IEEE PerCom 2020**), Austin, USA. pp. 1 - 8. DOI:10.1109/PerCom45495.2020.9127359.

[29] **Surjya Ghosh**, Kaustubh Hiware, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Does Emotion Influence the Use of Auto-suggest during Smartphone Typing?", 24th International Conference on Intelligent User Interfaces (**ACM IUI 2019**), Los Angeles, USA. pp. 144-149. DOI:10.1145/3301275.3302329.

[30] **Surjya Ghosh**, Shivam Goenka, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Representation Learning for Emotion Recognition from Smartphone Keyboard Interactions", 8th International Conference on Affective Computing & Intelligent Interaction (**ACII 2019**), Cambridge, UK. pp. 704-710. DOI:10.1109/ACII.2019.8925518.

[31] Rohit Verma, **Surjya Ghosh**, Saketh Mahankali, Niloy Ganguly, Bivas Mitra, and Sandip Chakraborty, "ComfRide: A Smartphone based System for Comfortable Public Transport Recommendation", 12th ACM Conference on Recommender Systems (**ACM RecSys 2018**), Vancouver, Canada. pp. 181-189. DOI:10.1145/3240323.3240359.

[32] **Surjya Ghosh**, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Evaluating Effectiveness of Smartphone Typing as an Indicator of User Emotion", 7th International Conference on Affective Computing and Intelligent Interaction (**ACII 2017**), San Antonio, Texas, USA. pp. 146-151. DOI:10.1109/ACII.2017.8273592.

[33] **Surjya Ghosh**, Niloy Ganguly, Bivas Mitra, and Pradipta De, "TapSense: Combining Self-Report Patterns and Typing Characteristics for Smartphone based Emotion Detection", 19th International Conference on Human-Computer Interaction with Mobile Devices and Services (**ACM MobileHCI 2017**), Vienna, Austria. pp. 1-12. DOI:10.1145/3098279.3098564.

[34] Rohit Verma, **Surjya Ghosh**, Niloy Ganguly, Bivas Mitra, and Sandip Chakraborty, "Smartphone based Spatio-temporal Sensing for Annotated Transit Map Generation", 25th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (**ACM SIGSPATIAL 2017**) California, USA. pp. 1-10. DOI:10.1145/3139958.3140005.

[35] **Surjya Ghosh**, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Towards Designing an Intelligent Experience Sampling Method for Emotion Detection", 14th Annual IEEE Consumer Communications & Networking Conference (**IEEE CCNC 2017**), Las Vegas, USA. pp. 401-406. DOI:10.1109/CCNC.2017.7983143.

[36] Rohit Verma, **Surjya Ghosh**, Aviral Shrivastava, Niloy Ganguly, Bivas Mitra, and Sandip Chakraborty, "Unsupervised Annotated City Traffic Map Generation", 24th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (**ACM SIGSPATIAL 2016**) California, USA. pp. 1-4. DOI:10.1145/2996913.2996942.

Articles in Peer-reviewed Conference (Workshops, Posters, Demos)

[37] Panyu Zhang, Shohruh Shokulov, Humoyunbek Abdukarimov, Jumabek Alikhanov **Surjya Ghosh**, and Uichin Lee, "Training-Free Few-Shot Personalization for Mobile Mood Inference with TabPFN", The 2nd DiversityOne Open Challenge at UbiComp/ISWC 2026 (**ACM UbiComp Companion'26**) Shanghai, China. DOI:10.1145/3798063.3842639.

[38] Shreyans Jain, Thomas Robinson, **Surjya Ghosh**, Timothy Pierson, and Sougata Sen, "CSIWatch: Detecting eating activity using CSI data from a smartwatch worn on the non-dominant hand", 3rd International Workshop on Pervasive Wireless Sensing and Edge Computing (**WiSense 2026**), Pisa, Italy (Workshop colocated with IEEE PerCom 2026). DOI:10.1109/PerComWorkshops68308.2026.11585336.

[39] Shreyans Jain, Yash Bhisikar, **Surjya Ghosh**, Timothy Pierson, and Sougata Sen, "SandDune: Single ANtenna Device for Detecting User's Natural Eating Habits", The 23rd International Conference on Pervasive Computing and Communications (**IEEE PerCom 2025**), Washington DC, USA (Work-in-progress). DOI:10.1109/PerComWorkshops65533.2025.00164.**(Runner Up Best WIP Paper)**

[40] Swarnali Banik, Akhilesh Adithya, Bivas Mitra, Sougata Sen, Snehanshu Saha, and **Surjya Ghosh**, "PResCon: Physiological Response Augmented Continuous Emotion Annotation Tool", 17th International Conference on COMmunication Systems & NETworkS (**COMSNETS 2025**), Bangalore, India. DOI:10.1109/COMSNETS63942.2025.10885663. (Demo)

[41] Hrishikesh Govindrao Kusneniwar, **Surjya Ghosh**, and Sougata Sen, "SonicGlass: An Obstacle Detection and Navigation System Using Smartglass-Based Ultrasonic Sensors", 16th International Conference on COMmunication Systems & NETworkS (**COMSNETS 2024**), Bangalore, India. DOI:10.1109/COMSNETS59351.2024.10427277. (Poster)

[42] Anuja Pinge, Dheryta Jaisinghani, **Surjya Ghosh**, Aditya Challa, and Sougata Sen, "mTanaaw: A System for Assessment and Analysis of Mental Health with Wearables", Workshop on Networking Humanitarian Technology for Healthcare at COMSNETS (**NetHealth 2024**), Bangalore, India. DOI:10.1109/COMSNETS59351.2024.10427432.

[43] Harman M. Singh, Shrishailya Agashe, Shreyans Jain, **Surjya Ghosh**, Aditya Challa, Sravan Danda, and Sougata Sen, "Insights from Executing TinyML Models on Smartphones and Microcontrollers", 19th Annual International Conference on Distributed Computing in Smart Systems and the Internet of Things (**DCOSS-IoT 2023**), Pafos, Cyprus. DOI:10.1109/DCOSS-IoT58021.2023.00026. (Poster)

[44] Saumya Mathkar, Prakhar Karsh, Udit Baluja, **Surjya Ghosh**, Sougata Sen, and Vinayak Naik , "A Smartphone-based Application to Detect Parkinson's Disease Using Audio", 15th International Conference on COMmunication Systems & NETworkS (**COMSNETS 2023**), Bangalore, India. DOI:10.1109/COMSNETS56262.2023.10041413. (Demo)

[45] Akhilesh Adithya, Snigdha Tiwari, Sougata Sen, Sandip Chakraborty, and **Surjya Ghosh**, "OCEAN: Towards Developing an Opportunistic Continuous Emotion Annotation Framework", The 20th International Conference on Pervasive Computing and Communications (**IEEE PerCom 2022**), Pisa, Italy (Work-in-progress). DOI:10.1109/PerComWorkshops53856.2022.9767359.

[46] Anuja Pinge, Soumyadip Bandyopadhyay, **Surjya Ghosh**, and Sougata Sen, "A Comparative Study between ECG-based and PPG-based Heart Rate Monitors for Stress Detection", Workshop on Networking Humanitarian Technology for Healthcare at COMSNETS (**NetHealth 2022**), Bangalore, India. pp 84 - 89. DOI:10.1109/COMSNETS53615.2022.9668342.

[47] **Surjya Ghosh**, Sumit Sahu, Niloy Ganguly, Bivas Mitra, and Pradipta De, "EmoKey: An Emotion-aware Smartphone Keyboard for Mental Health Monitoring", 11th International Conference on Communication Systems and Networks (**COMSNETS 2019**), Bangalore, India. pp. 496-499. DOI:10.1109/COMSNETS.2019.8711078. (Poster) **(Best Paper Award - 3rd Place)**

[48] Suman Kalyan Maity, Ankan Mullick, **Surjya Ghosh**, Anil Kumar, Sunny Dhamnani, Sudhansu Bahety, and Animesh Mukherjee, "Understanding Psycholinguistic Behavior of Predominant Drunk Texters in Social Media", In IEEE ISCC Workshops - ICTS4eHealth (**IEEE ICTS4eHealth 2018**) Natal, Brazil. pp. 01096-01101. DOI:10.1109/ISCC.2018.8538637. **(Best Paper Award)**

[49] **Surjya Ghosh**, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Effectiveness of Deep Neural Network Model in Typing-based Emotion Detection on Smartphones", 24th Annual International Conference on Mobile Computing and Networking (**ACM Mobicom 2018**), New Delhi, India. pp. 750-752. DOI:10.1145/3241539.3267761. (Poster)

[50] **Surjya Ghosh**, "Emotion-aware Computing using Smartphone", 9th International Conference on COMmunication Systems & NETworkS (**COMSNETS 2017**), Bangalore, India. pp. 592-593. DOI:10.1109/COMSNETS.2017.7945464. (Graduate Forum)

[51] **Surjya Ghosh**, Vatsalya Chauhan, Niloy Ganguly, Bivas Mitra, and Pradipta De, "Impact of Experience Sampling Methods on Tap Pattern based Emotion Recognition", 4th ACM Workshop on Mobile Systems for Computational Social Science - MCSS (**ACM UbiComp/ISWC.15 Adj**) Osaka, Japan. pp. 713-722. DOI:10.1145/2800835.2804396.

Sponsored Projects

- *GameSignAI: Using AI to Help Deaf Children Acquire Language Skills*
  – Sponsor: NSF-DST, Dept. of Science and Technology (DST), Govt. of India.
  – Role: Co-Principal Investigator.
  – Duration: 2025 - 2027. (Ongoing)

- *HEXR: Hybrid Explainable Robust Learning Framework to Reduce Annotation Overhead*
  – Sponsor: SERB (Science & Engineering Research Board), Dept. of Science and Technology (DST), Govt. of India.
  – Role: Principal Investigator.
  – Duration: 2023 - 2026. (Ongoing)

- *e-Monitor: Development of AI driven Smartphone based Assistive Framework to Monitor Student Stress as well as Learning Engagements in Online Learning Platform*
  – Sponsor: AI4ICPS, IIT Kharagpur, India
  – Role: Investigator.
  – Duration: 2023 - 2025. (Ongoing)

- *CDRF: Separation of Carcinogenic Industrial Solvents Using Solar Energy, Encapsulated PCM based Magnetic Nanoparticles and AI applications for Optimization: Towards a Sustainable Alternative*
  – Sponsor: BITS Pilani Goa, India.
  – Role: Principal Investigator.
  – Duration: 2023 - 2025. (Completed)

- *EDGESYS End-to-end Framework for Edge Systems*
  – Sponsor: BITS Pilani Goa, India.
  – Role: Principal Investigator.
  – Duration: 2022 - 2024. (Completed)

- *SMARTHEALTH: Smart Device based Health Monitoring*
  – Sponsor: BITS Pilani Goa, India.
  – Role: Principal Investigator.
  – Duration: 2021 - 2023. (Completed)

Industry Collaboration

- *Designing Multi-modal Affect Detection Methodology* Jul 2019 - Jul 2021
  – The goal of the project is to develop a multi-modal emotion inference technology for a leading automotive organization in Europe. I worked in this project as a postdoctoral researcher to design prototype, perform user studies, collect dataset, develop AI models, and validate the proposed solution using quantitative and qualitative approaches.

- *Behavior Modeling in Multi-sensor Environments* Aug 2018 - Jul 2019
  – The objective of this project was to leverage different sensor data from the environment for behavior modeling. This was a collaborative project with Indian tech giant TCS (TATA Consultancy Services Ltd.). My role in the project was to investigate the smartphone and wearable usage logs (app usage details, physiological signals) for mental state detection applying state-of-the-art ML models.

PhD Supervision

PhD Scholar	Thesis Topic	Status	Duration
Swarnali Banik	Developing Intelligent Annotation Approaches for Fine-grain, Continuous Emotion Labeling in Video Applications	Submitted	Aug 2022 – Jun 2026
Chandani Jha	Towards Addressing Interpersonal Differences in Physiological Sensing-based Emotion Engagement Detection	Ongoing	Jan 2026 – Present

Teaching Experience

- Instructor, BITS Pilani, Goa, India

Sl.	Course	Semester and Academic Year
1	Pervasive Computing	Semester I (Autumn): 2021–2022, 2022–2023, 2023–2024, 2024–2025, 2025–2026, 2026–2027
2	Human–Computer Interaction	Semester II (Spring): 2021–2022, 2022–2023, 2024–2025, 2025–2026
3	Operating Systems	Semester I (Autumn): 2025–2026
4	Computer Programming	Semester II (Spring): 2023–2024
5	Computer Programming Lab	Semester II (Spring): 2023–2024, 2024–2025
6	Computer Architecture Lab	Semester I (Autumn): 2023–2024
7	Operating Systems Lab	Semester II (Spring): 2022–2023
8	Computer Networks Lab	Semester I (Autumn): 2021–2022

Scientific Community Service

- Editorial Board Member
  – Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (**IMWUT**) Mar'2026 - till date.
- Journal Reviewing
  – ACM Transactions on Computing for Healthcare (2025), IEEE Transactions on Affective Computing (TAFFC) 2024, IEEE Transactions on Multimedia (TMM) 2020, Elsevier Ad Hoc Networks 2021, PACM IMWUT (2024, 2021, 2020), ACM ISS 2022.
- Conference Reviewing
  – ACM CHI (2025, 2022, 2021, 2020), ACM ICMI (2023), ACII (2022,2021), ACM Automotive UI 2021, ACM MobileHCI 2020, AffCon workshop, co-located with AAAI 2019.
- Senior PC Member
  – ACM ICMI (2025, 2024).

- TPC Member
  – ACII (2025,2024, 2023), IEEE AIVR 2020, COMSNETS (Graduate Forum, Posters) 2022, CODS-COMAD (Young Researcher Symposium) 2022.
  – ACM S^3 workshop, in conjunction with Mobicom 2018.

- Organising Committee Member
  – Graduate forum chair, IndoML 2024.
  – Invited jury member at KPIT Sparkle 2024.
  – Poster chair, ICDCN 2023; Publicity chair, COMSNETS 2023.
  – Student Consortium chair, India HCI 2022.
  – Student volunteer, ACM IUI 2019.

Invited Talks

This list does not contain the conference talks.

- 13th Indo-German Frontiers of Engineering Symposium, Mumbai, India Dec, 2024
  – *Title: Investor Behavior Modeling in Financial Market Using Digital Twin*

- KAIST, Republic of Korea Oct, 2024
  – *Title: PResUP: Physiological Response based User Profiling for Opportunistic Emotion Annotation*

- PCCE College, Goa, India Jan, 2023
  – *Title: Recent trends in Machine Learning, Computer Vision and Data Science*

- Leibniz AI Lab, Germany Jun, 2021
  – *Title: Developing Smartphone Keyboard Interaction based Emotion Detection System*

- Dutch CHI 2020, Netherlands Jun, 2020
  – *Title: Towards Improving Emotion Self-report Collection using Self-reflection*

- India HCI 2019, Hyderabad, India Nov, 2019
  – *Title: Does Emotion Influence the Use of Auto-suggest during Smartphone Typing?*

- Cornell Tech, New York City, USA Mar, 2019
  – *Title: Smartphone-based emotion detection: Research Challenges, System Implementation, and Applications*

Travel Grants and Awards

- Received ACM/IARCS Travel Grant for attending ACM CHI 2024.
- Received Microsoft Research India Travel Grant for attending IUI 2019, MobileHCI 2017, UbiComp 2015.
- LRN India Travel Grant for attending IUI 2019.
- Received Student Travel Grant for attending IUI 2019, Mobicom 2018, and ACII 2017.
- Received COMSNETS Travel Grant for attending COMSNETS 2017, 2018, 2019.
- Ranked 10th in merit panel of Assistant Professor at WBCSC in July 2016.
- Qualified in UGC-NET Exam (Computer Science) for Assistant Professor in Jun 2014 among $\approx 50,000$ candidates (qualify rate $\approx 5\%$) across India.